Matplotlib – 5 Practical Questions

Question 1: Line Chart

Dataset: Monthly Sales

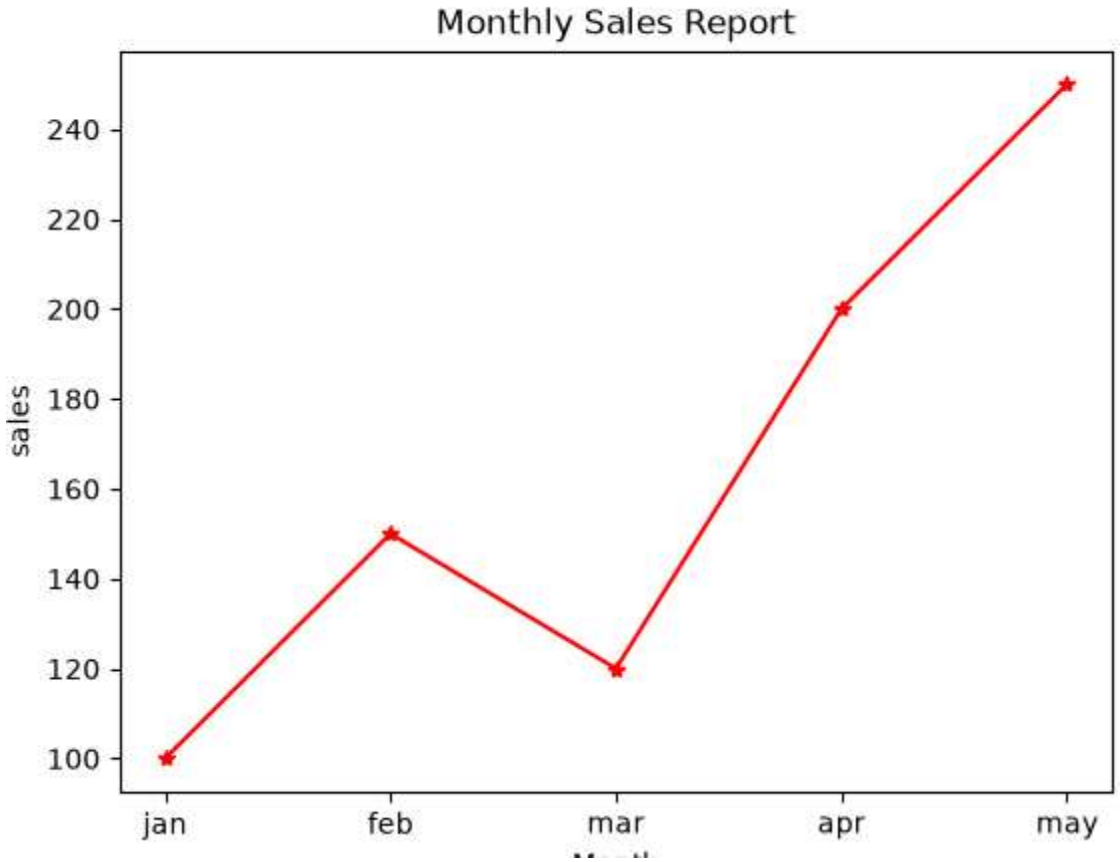
Month	Sales
Jan	100
Feb	150
Mar	120
Apr	200
May	250

Task: Create a line chart with the following features:

- Line color must be red.
- Add markers to each data point.
- X-axis label: `Month`
- Y-axis label: `Sales`
- Title: `Monthly Sales Report`

```
C:\Windows\System32\cmd.exe - python
[notice] To update, run: python.exe -m pip install --upgrade pip

(env13) D:\STUDENTS\HARSHINI PRASANNA\NUMPY>python
Python 3.11.2 (tags/v3.11.2:878ead1, Feb  7 2023, 16:38:35) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> import matplotlib.pyplot as plt
>>>
>>> x=['jan' ,'feb' ,'mar' ,'apr' ,'may' ]
>>> y=[ 100 , 150 ,120 ,200 , 250]
>>>
>>> plt.plot( x , y  ,linestyle= "-", marker = "*", color="red")
[<matplotlib.lines.Line2D object at 0x000001E63220E690>]
>>> plt.xlabel("Month" )
Text(0.5, 0, 'Month')
>>> plt.ylabel("sales" )
Text(0, 0.5, 'sales')
>>> plt.title("Monthly Sales Report")
Text(0.5, 1.0, 'Monthly Sales Report')
>>> plt.show()
>>>
```



Question 2: Scatter Plot

Dataset: Student Marks

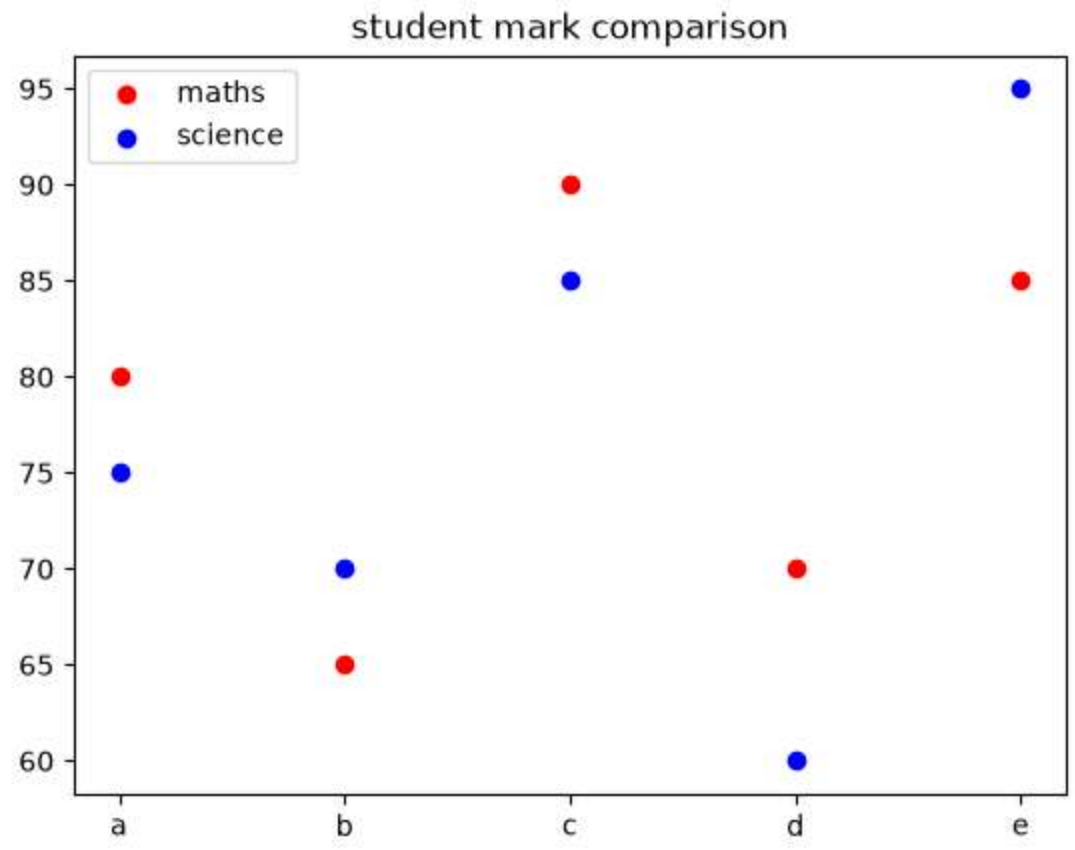
Student	Maths	Science
A	80	75
B	65	70
C	90	85
D	70	60
E	85	95

Task: Create a scatter plot with the following features:

- Plot Maths marks in blue and Science marks in red.
- Use different markers for each subject.
- Add legends named `Maths` and `Science`.
- Title: `Student Marks Comparison`

C:\Windows\System32\cmd.exe - python

```
>>>
>>>
>>> import matplotlib.pyplot as plt
>>> x=['a','b','c','d','e']
>>> y=[80,65,90,70,85]
>>> z=[75,70,85,60,95]
>>> plt.scatter (x, y , marker="o",color="red", label="maths")
<matplotlib.collections.PathCollection object at 0x000001E632357CD0>
>>> plt.scatter (x, z , marker="o",color="blue",label="science")
<matplotlib.collections.PathCollection object at 0x000001E636147D10>
>>>
>>> plt.legend()
<matplotlib.legend.Legend object at 0x000001E63612AF50>
>>> plt.title("student mark comparison")
Text(0.5, 1.0, 'student mark comparison')
>>> plt.show()
```



Question 3: Bar Chart

Dataset: Product Sales

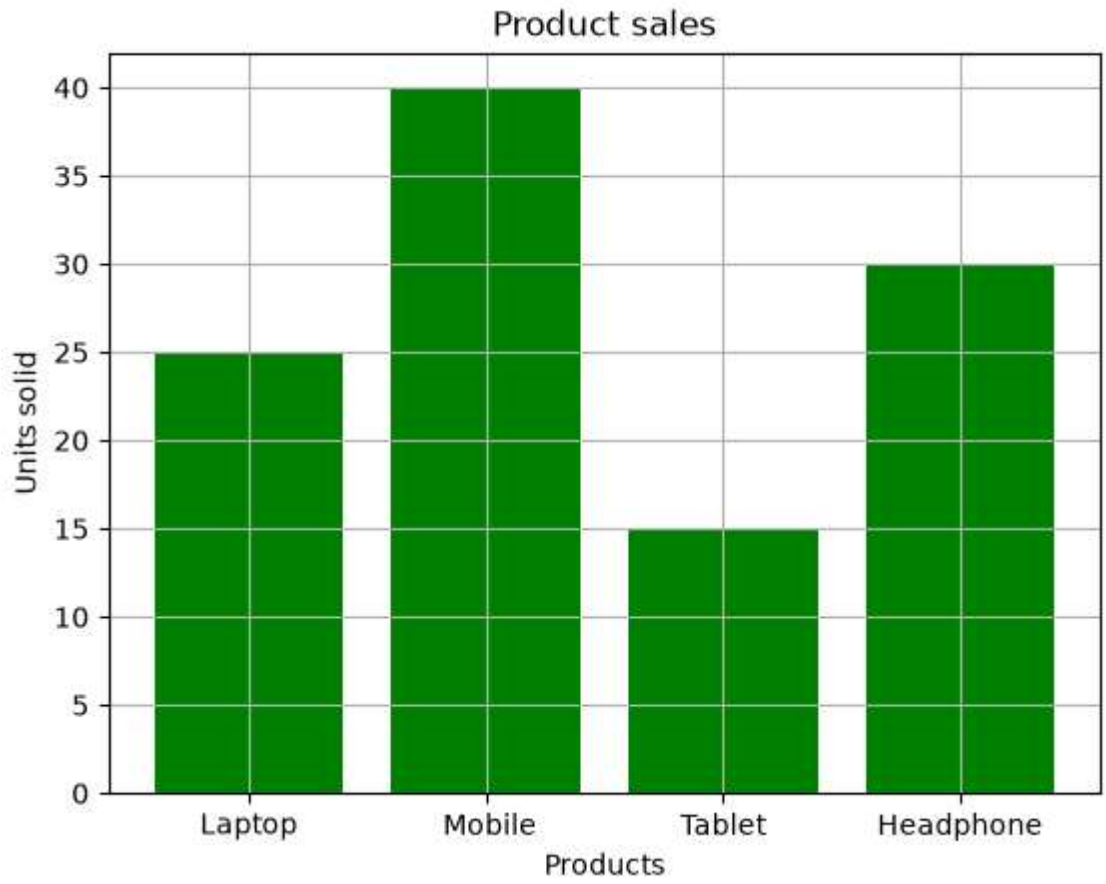
Product	Units Sold
Laptop	25
Mobile	40
Tablet	15
Headphones	30

Task: Create a bar chart with the following features:

- Bars must be green.
- Add the title `Product Sales`.
- Label the X-axis as `Products` and Y-axis as `Units Sold`.
- Display gridlines on the Y-axis only.

C:\Windows\System32\cmd.exe - python

```
>>>
>>> import Matplotlib.pyplot as plt
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ModuleNotFoundError: No module named 'Matplotlib'
>>>
>>> Products=[ "Laptop" ," Mobile" , "Tablet" , "Headphone" ]
>>> Units_solid=[ 25 , 40 , 15 , 30 ]
>>> plt.bar (Products, Units_solid,color= "green")
<BarContainer object of 4 artists>
>>> plt.xlabel("Products")
Text(0.5, 0, 'Products')
>>> plt.ylabel("Units solid")
Text(0, 0.5, 'Units solid')
>>> plt.title("Product sales")
Text(0.5, 1.0, 'Product sales')
>>> plt.grid(True)
>>> plt.show()
>>>
```



Question 4: Histogram

Dataset: Student Exam Scores

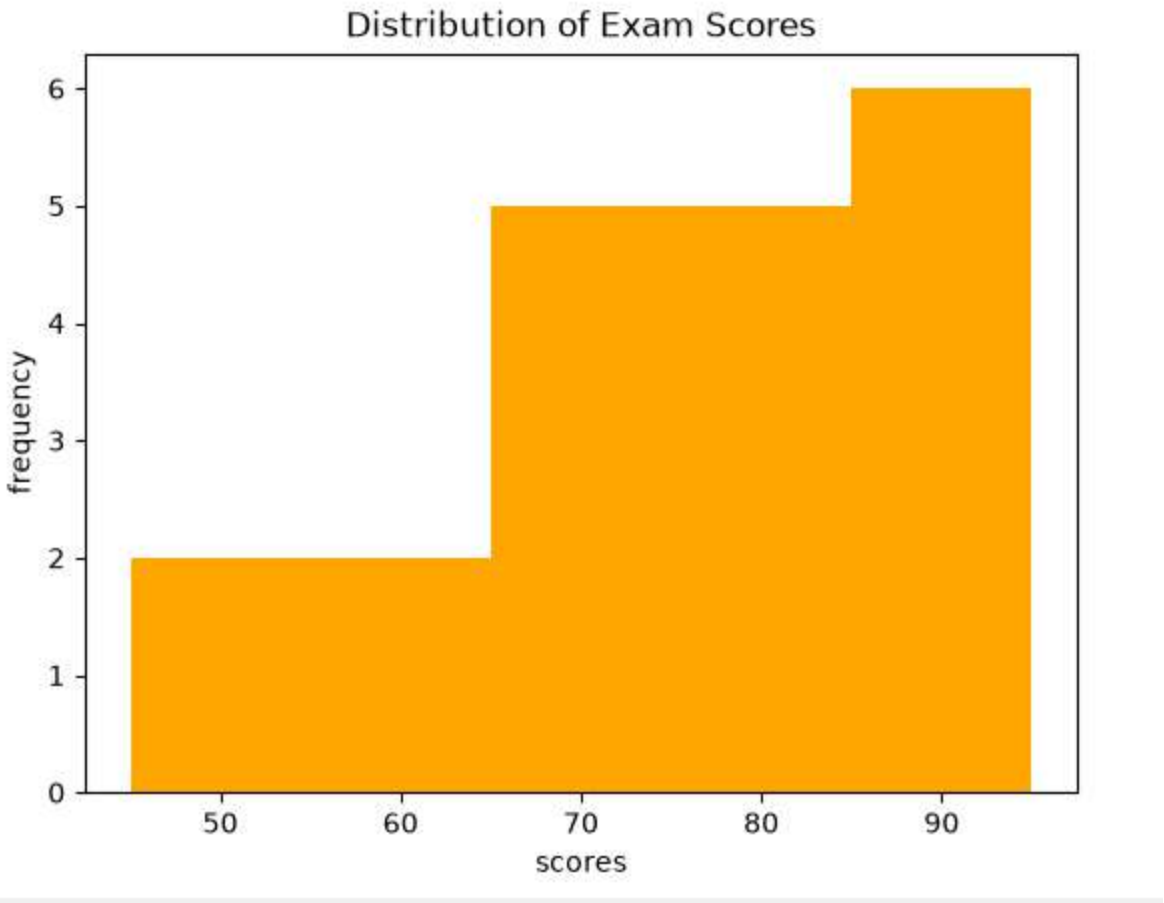
scores = [45, 55, 60, 65, 70, 72, 75, 80, 82, 85, 88, 90, 92, 95, 50, 68, 73, 78, 84, 89]

Task: Create a histogram with the following features:

- Use 5 bins.
- Set the bar color to orange.
- Add the title `Distribution of Exam Scores`.
- Label the X-axis as `Scores` and Y-axis as `Frequency`.

C:\Windows\System32\cmd.exe - python

```
>>>
>>> import Matplotlib.pyplot as plt
Traceback (most recent call last):
  File "<stdin>", line 0, in <module>
KeyboardInterrupt
>>>
>>> scores = [45, 55, 60, 65, 70, 72, 75,80, 82, 85, 88, 90, 92, 95, 50, 68, 73,78, 84, 89]
>>> plt.hist( scores , bins=5,color= "orange")
(array([2., 2., 5., 5., 6.]), array([45., 55., 65., 75., 85., 95.]), <BarContainer object of 5 artists>)
>>> plt.xlabel("scores")
Text(0.5, 0, 'scores')
>>> plt.ylabel("frequency")
Text(0, 0.5, 'frequency')
>>> plt.title("Distribution of Exam Scores")
Text(0.5, 1.0, 'Distribution of Exam Scores')
>>> plt.show()
>>>
```



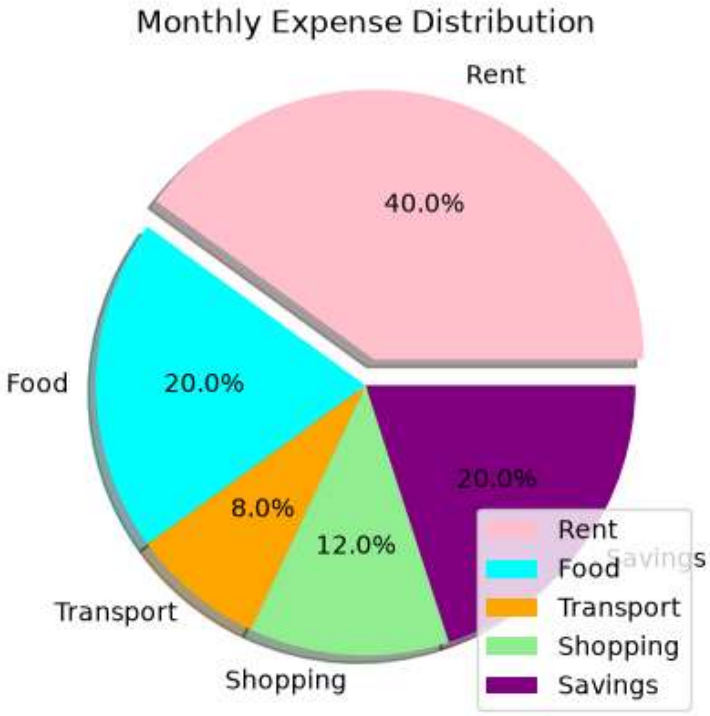
Question 5: Pie Chart

Dataset: Monthly Expenses

Category	Amount (₹)
Rent	10000
Food	5000
Transport	2000
Shopping	3000
Savings	5000

Task: Create a pie chart with the following features:

- Display category names as labels.
- Show percentages on each slice.
- Explode the `Rent` slice.
- Add a shadow effect.
- Add the title `Monthly Expense Distribution`.



C:\Windows\System32\cmd.exe - python

```
>>>
>>> import matplotlib.pyplot as plt
>>>
>>> category = ["Rent", "Food", "Transport", "Shopping", "Savings"]
>>> amount = [10000, 5000, 2000, 3000, 5000]
>>>
>>> explode = (0.1, 0, 0, 0, 0)
>>> colors = ["pink", "cyan", "orange", "lightgreen", "purple"]
>>>
>>> plt.pie(
...     amount,
...     labels=category,
...     explode=explode,
...     colors=colors,
...     autopct='%1.1f%%',
...     shadow=True
... )
<matplotlib.container.PieContainer object at 0x000001E632306490>
>>>
>>> plt.legend()
<matplotlib.legend.Legend object at 0x000001E63235F450>
>>> plt.title("Monthly Expense Distribution")
Text(0.5, 1.0, 'Monthly Expense Distribution')
>>> plt.show()
```